WAHENGBAM HERAMANI SINGH

B.Tech CSE Student & AI-Oriented Developer | NLP • LLMs • RAG • Full-Stack Systems

Phone | Email | GitHub | LinkedIn | Portfolio
NIT Manipur, Imphal, India

PROFESSIONAL SUMMARY

Computer Science undergraduate at NIT Manipur building AI-assisted systems and real-time web applications. Interested in Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and applied AI for social impact. Hands-on experience developing full-stack applications and backend systems using Python, Django, React, WebSockets, Redis, PostgreSQL, and Docker.

EDUCATION

National Institute of Technology Manipur — B.Tech in Computer Science & Engineering | 3rd Semester | **CGPA: 8.33/10** | 2025–2029 (Ongoing)
Relevant Coursework: Data Structures, Algorithms, Discrete Mathematics, Digital Logic Design,Programming in Python, Mathematics for Computing,Managerial Economics.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, TypeScript, SQL
Frameworks & Tools: Django, Django REST Framework, React, WebSockets, Celery, Redis, Docker, Nginx, PostgreSQL, Git, Linux VPS
AI/ML & Concepts: RAG, Ollama, Qdrant, LLM grounding, local LLMs (Llama, Qwen), embeddings, constraint reasoning

PROJECTS & TECHNICAL EXPERIENCE

HOUSIE — Real-Time Multiplayer Gaming Platform | **Lead Architect** • Django Channels, WebSockets, Redis, Celery, React, Docker **| 2026**

- Architected and deployed an asynchronous multiplayer backend using Django Channels, WebSockets, and Redis pub/sub.
- Containerized the platform with Docker and managed background jobs with Celery for real-time game operations.
- Implemented player/admin workflows, automated number calling, prize validation, and Nginx reverse proxy with SSL on a Linux VPS.

PAANGBA — AI-Assisted Financial Access & Lender-Matching Platform | Contributor/Lead Research **|Present-2026**

- Working on a platform designed to help citizens seeking financial assistance discover relevant private-lender opportunities through consent-based information sharing.
- Contributing to AI-assisted document checks, potential forgery and information-mismatch detection, and the generation of qualified leads for participating lenders.
- The platform aims to improve the quality of applicant information and help lenders assess relevant enquiries more efficiently, while keeping data sharing consent-driven.

Coaching Website — Serverless Content Management Platform | Developer | Live Demo **| 2026**

- Google Apps Script, Google Drive, Google Sheets, JavaScrip
- Developed a coaching website where students access learning materials and teachers upload and manage PDFs, images, and educational resources.
- Integrated Google Apps Script with Google Drive for file storage and Google Sheets for structured data and activity logging, leveraging existing Google infrastructure to avoid separate hosting and paid database costs.

AI-Driven Scheme Matching for Marginalized Entrepreneurs | Research Lead • Python, LLMs, RAG, Information Retrieval **| SIH-2026**

- Designed an eligibility-matching approach combining RAG with structured constraint logic to generate grounded recommendations for government assistance schemes.
- Developed a verification-oriented retrieval concept mapping scheme requirements to applicant profiles; selected as Team Leader at the NIT Manipur internal hackathon.

Applied LLM & RAG Pipeline Architecture Exploration | Applied Research • Ollama, Llama, Qwen, Qdrant, Embeddings **| 2026**

- Explored local LLM inference and RAG pipelines; implemented dense embeddings, semantic search, and context-grounding checks to reduce unsupported outputs.
- Administered Linux services involving containers, reverse proxies, caching, and task queues, with practical exposure to network security.

LEADERSHIP & KEY ACHIEVEMENTS

- **AI for Seva Hackathon — 5th Place, Team Leader:** Led the team to 5th place among 52 teams; organised by the Department of Information Technology, Government of Manipur.
- **Smart India Hackathon (SIH) 2026 — Team Leader:** Led the team through NIT Manipur's internal selection, qualifying for the national stage.
- **Department Newsletter Student Batch Lead (Sep 2026):** Coordinated achievement records and publications for the CSE Department at NIT Manipur.
- **Shraman Foundation Scholarship (2026):** Selected undergraduate scholar recognised for academic excellence and technical potential.

ACTIVITIES & LANGUAGES

Sports & Activities: Football, Badminton
Languages: English (Professional Working Proficiency), Manipuri/Meitei (Native), Hindi (Working Proficiency).